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Machine learning-based prognostic prediction for acute ischemic stroke using whole-brain and infarct multi-PLD ASL radiomics

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Abstract

Introduction Accurate early prognostic prediction for acute ischemic stroke (AIS) is essential for guiding personalized treatment. This study aimed to assess the predictive value of radiomics features from whole-brain and infarct cerebral blood flow (CBF) images using multiple post-labeling delay arterial spin labeling (multi-PLD ASL), and to develop a prediction model incorporating clinical risk factors.

Methods Radiomics features were extracted from the whole-brain and infarct regions based on multi-PLD ASL CBF images of 110 AIS patients. Five machine learning algorithms were used to construct radiomics models (whole-brain, infarct, and combined), clinical models, and comprehensive models integrating radiomics and clinical data. Model performance and clinical utility were assessed using receiver operating characteristic and decision curve analyses. Model stability was evaluated via 5000 permutation tests, and differences in the area under the curve (AUC) were compared using the DeLong test. Shapley Additive exPlanation was used to interpret feature contributions.

Results The whole-brain and infarct radiomics models showed similar predictive performance. The combined radiomics models generally outperformed the infarct-only models. Additionally, no significant differences were observed between the combined radiomics models and the clinical models across the five algorithms. The comprehensive models, which integrated both radiomics and clinical features, demonstrated superior performance compared to both the clinical and combined radiomics models. Among all models, the comprehensive model based on a support vector machine achieved the highest predictive performance (AUC = 0.904). Its predictive capability was primarily driven by the baseline National Institutes of Health Stroke Scale score, age, infarct shape features, and higher-order statistical and texture features derived from both infarct and whole-brain CBF images.

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Conclusions The whole-brain CBF radiomics features of multi-PLD ASL can replace the infarct radiomics features that need to be delineated. Combined radiomics models outperformed infarct-only models and performed similarly to clinical models, which is applicable to AIS patients with incomplete clinical data. The comprehensive model integrates multi-PLD ASL CBF radiomics and clinical data, providing a safe and accurate tool for early prognosis prediction in AIS patients.

Keywords Acute ischemic stroke, Radiomics, Arterial spin labeling, Cerebral blood flow, Machine learning

Introduction

Acute ischemic stroke (AIS) refers to a series of clinical syndromes caused by embolism of local cerebral arteries leading to neurological deficits [1]. It is estimated that by 2030, the global age-standardized incidence rate of AIS will continuously increase to 89.32 per 100,000 people [2]. Despite receiving effective treatment, around 20–40% of AIS patients still suffer from permanent disability [3]. This affects both the quality of life of patients and imposes a significant economic burden on society [4]. As the global population ages, the pressure on public health from AIS is anticipated to rise further [5]. In the early stages of AIS, clinicians can implement aggressive treatment strategies to improve patient prognosis. Therefore, identifying potential AIS patients with an unfavorable prognosis at an early stage has become a crucial focus in the clinic.

Cerebral blood flow (CBF) serves as a significant prognostic indicator for AIS patients [6]. Currently, the assessment of CBF primarily relies on dynamic susceptibility contrast perfusion weighted imaging (DSC-PWI) or computed tomography perfusion images [7]. However, these technologies rely on the use of gadolinium or iodine contrast agents, which are not only expensive but also carry potential risks of allergic reactions and kidney damage, thus limiting their widespread clinical application [8].

Arterial spin labeling (ASL) has become increasingly important for evaluating CBF in recent years as a non-invasive magnetic resonance imaging (MRI) technique that does not require exogenous contrast agents [9]. In particular, the multiple post-labeling delay (PLD) ASL technique is able to encode different PLD in a single scan and obtain CBF corrected by the varying arterial transit time (ATT) in brain tissue [10]. This significantly enhances the systemic underestimation of CBF linked with single-PLD methods. In general, the ATT in ischemic regions of AIS patients is notably prolonged [11]. Therefore, this correction is especially crucial. Several studies have shown that multi-PLD ASL, which has been widely used in neurological research, provides highly consistent CBF measurements with DSC-PWI [12, 13].

Conventional visual assessment or quantitative analysis of CBF images is subject to the observer's subjective experience and does not capture the variability of lesion perfusion. Radiomics, a recently developed quantitative analysis technique, utilizes computer technology to extract numerous features from medical images that are visually difficult to detect, enabling a more objective analysis of heterogeneity [14]. Several studies have found that infarct radiomics features from DSC-PWI images can be used to predict the prognosis of AIS [15]. Through our earlier investigation utilizing multi-PLD ASL technology, it was determined that the significance of infarct CBF radiomics features for AIS prognosis was considerable [16]. However, the impact of AIS extends beyond local CBF abnormalities, potentially involving other regions of the brain or even the entire brain [17]. Thamm et al. suggested that CBF in the unaffected hemisphere of AIS patients is a significant predictor of positive neurological outcomes, as the healthy hemisphere may provide collateral blood flow to the injured region of the opposite hemisphere [18]. Meanwhile, Guo et al. discovered that ischemia can be compensated for by the circulatory system of the contralateral hemisphere via the circle of Willis [19]. Additionally, acute ischemic injury can cause blood-brain barrier breakdown, causing irregular blood flow throughout the entire brain [20]. Consequently, there is an increasing curiosity in studying radiomics features of CBF images throughout the whole brain.

AIS prognostic prediction has been successfully achieved through the use of whole-brain radiomics features from DSC-PWI images in recent studies [21–23]. Additionally, Guo et al. found that whole-brain radiomics features outperformed infarct radiomics features in predictive accuracy and exhibited a certain synergistic effect when combined [24]. Therefore, it is crucial to explore their combined influence in predicting AIS prognosis. Nevertheless, the prognostic potential of radiomics derived from ASL has not been thoroughly investigated. Given its non-invasive nature and ability to quantify CBF without the use of contrast agents, ASL represents a promising yet underutilized source of prognostic imaging biomarkers.

Therefore, this study aims to explore the predictive value of whole-brain and infarct CBF radiomics features for clinical prognosis in AIS patients through multi-PLD ASL technology. The objective is to establish a model capable of early predicting AIS prognosis; and offer a secure and economical prognostic assessment tool for clinics.

Methods

Patients

This study collected data from 110 AIS patients (63 males, 47 females) at Yancheng Third People's Hospital between December 2020 and October 2024. The participants' ages ranged from 40 to 92 years, with a mean age of 67.68 ± 12.32 years. Figure 1 illustrates the process of inclusion and exclusion.

Inclusion criteria: [1] diagnosis of AIS according to the guidelines [25]; [2] age ≥ 18 years; [3] head MRI performed within 24–72 h after onset; and [4] pre-stroke modified Rankin Scale (mRS) score of 0.

Exclusion criteria: [1] posterior circulation AIS; [2] presence of intracranial space-occupying lesions; [3] other neuropsychiatric disorders; [4] serious diseases of other systems; and [5] significant MRI artifacts.

Clinical assessment

Demographic and clinical data were collected from all participants, including age, gender, onset to therapy time, baseline National Institutes of Health Stroke Scale (NIHSS) score, treatment method, history of atrial fibrillation, hypertension, diabetes, hyperlipidemia, coronary heart disease, smoking, alcohol abuse, and history of stroke. Each participant was followed up by telephone at 90 days post-onset to assess the mRS score and was classified into the favorable prognosis group (mRS ≤ 2) or the unfavorable prognosis group (mRS > 2).

MRI acquisition

All participants underwent head MRI scanning (3.0T, GE Discovery 750 W, USA), with sequences including axial three-dimensional T1-weighted imaging (3D-T1WI, TR: 7.5 ms, TE: 2.8 ms), axial diffusion weighted imaging (DWI, TR: 3586 ms, TE: 77.2 ms), and multi-PLD ASL (TR: 5978 ms; TE: 11.5 ms; FOV: 22 × 22 cm; slice thickness: 4.5 mm; slice number: 106; resolution: 4.67 mm * 4.67 mm; NEX: 1; PLD: 1.0 s, 1.22 s, 1.48 s, 1.78 s, 2.1 s, 2.63 s, 3.32 s; scan time: 6 min 2 s). The CBF images corrected for ATT were generated by averaging the individual CBF maps calculated for each PLD on the MRI scanner.

Imaging process

To enhance image quality, N4 bias field correction was first applied to all images to minimize the effects of

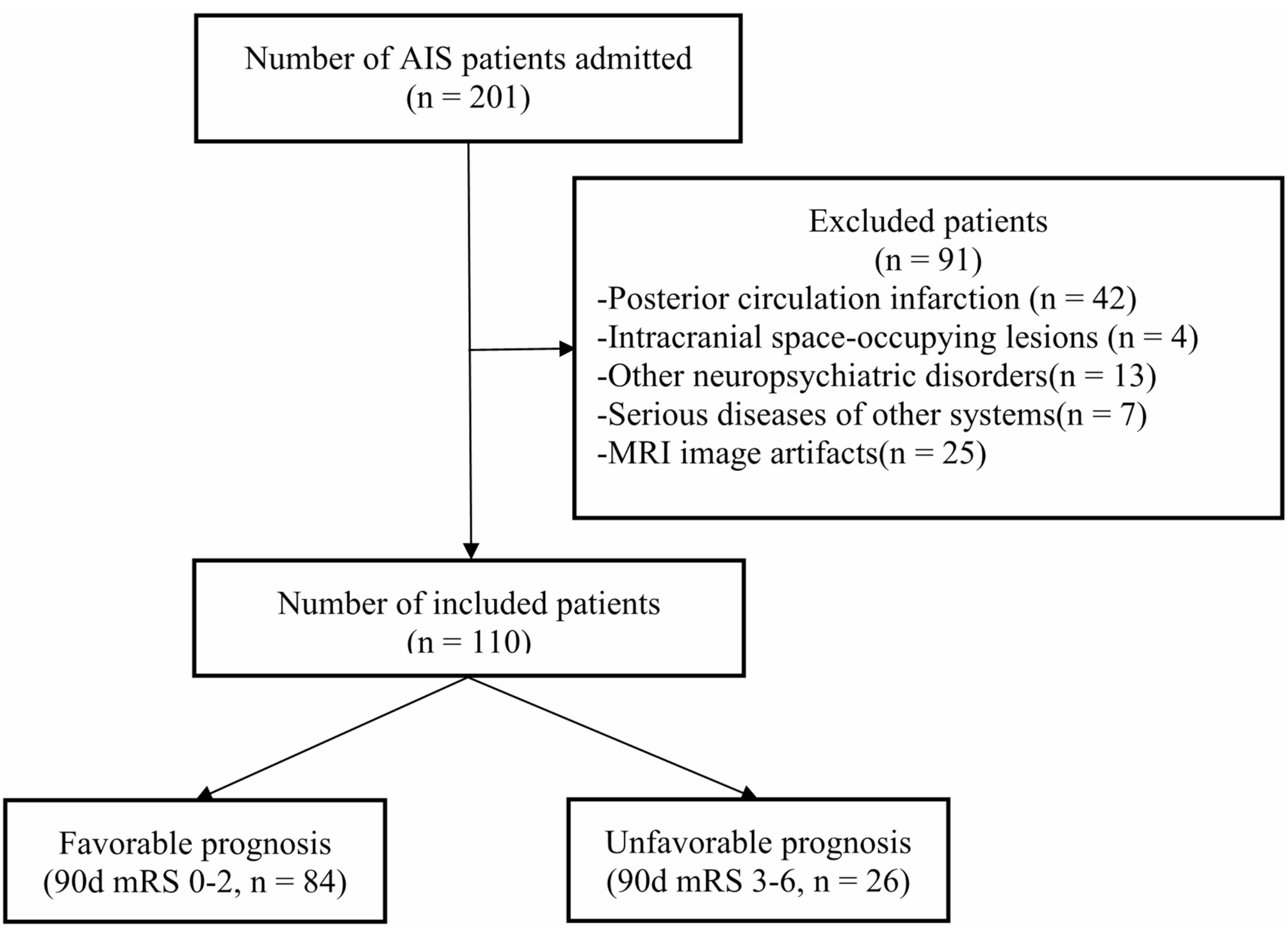

Fig. 1 Study enrollment process. AIS: acute ischemic stroke; MRI: magnetic resonance imaging; mRS: modified rankin scale

magnetic field inhomogeneity. Voxel-wise z-score normalization was then performed on the CBF maps within the whole-brain mask to reduce inter-individual variability in absolute perfusion values. Subsequently, the DWI images were utilized as a template for aligning with the CBF images. Two neuroradiologists delineated infarct as regions of interest (ROI) on the DWI images independently using ITK-SNAP (http://www.itk-snap.org, version 4.0), manually outlining them layer by layer. The delineated ROIs were subsequently moved to the registered CBF images. ROI selection was carefully done in order to avoid major blood vessels, sulci, and brain ventricles as much as possible.

Additionally, the standard tissue probability templates from SPM12 (http://www.fil.ion.ucl.ac.uk/spm) were used to segment gray matter, white matter, and cerebrospinal fluid regions from the 3D-T1WI images. A whole-brain mask was created from the segmentation results, covering gray matter, white matter, and cerebrospinal fluid. The mask was then applied to the CBF images after registration with the 3D-T1WI images.

Radiomics feature extraction

PyRadiomics software (http://pyradiomics.readthedocs.io, version 3.1.0) was used to extract 1132 radiomics features separately from both the infarct ROIs and whole-brain masks in the CBF images. These features encompass 14 shape features, 18 first-order statistics, 68 second-order texture features, and 1032 higher-order features (688 wavelet transform features and 344 Log filter features). Additionally, the interclass correlation coefficient (ICC) was used to assess the consistency of features extracted from the infarct ROIs in two separate delineations. Features with an ICC exceeding 0.75 were deemed to possess high reproducibility. Subsequently, both whole-brain and highly consistent infarct radiomics features were standardized to eliminate dimensional differences before being incorporated into the subsequent model training process.

Model building and evaluation

Five different machine learning (ML) methods were employed to construct the models, including logistic regression (LR), support vector machine (SVM), random forest (RF), extreme gradient boosting (XGBoost), and light gradient boosting machine (LightGBM). With the limited sample size in this study, leave-one-out cross-validation was chosen for model training over separating the subjects into training and testing sets. This training strategy iterates N (sample size) times, using (N – 1) samples for training in each iteration and the remaining one sample as the testing set. The model's performance was assessed based on the average prediction accuracy across all iterations. This method improves data utilization and prediction accuracy, making it well-suited for studies with limited samples [26]. Least absolute shrinkage and selection operator regression with 10-fold cross-validation was utilized for feature selection in each iteration. In order to enhance model performance, a grid search was carried out to systematically investigate various combinations of model-specific hyperparameters. Examples include the penalty parameter for LR, the kernel type and penalty parameter for SVM, and the number of estimators and maximum depth for RF, XGBoost, and LightGBM. Final hyperparameter settings were determined by evaluating the model performance metrics.

Infarct radiomics models, whole-brain radiomics models, and combined radiomics models (integrating both infarct and whole-brain radiomics features) were constructed. Additionally, continuous variables from clinical data (age, onset to therapy time, and baseline NIHSS score) were standardized, and clinical models were developed using the same model training approach. Ultimately, comprehensive models were constructed by integrating infarct and whole-brain radiomics features with clinical data. Receiver operating characteristic curves were generated for each model, and their predictive performance was evaluated based on area under the curve (AUC), sensitivity, specificity, accuracy, and F1-score. The bootstrap method was applied with 1,000 resampling iterations to estimate the 95% confidence interval for the AUC. Furthermore, it was observed that the proportion of patients with a favorable prognosis was significantly higher than that of those with an unfavorable prognosis. To further assess the model's performance in predicting the minority class, we additionally calculated the precision-recall AUC for the best-performing model in predicting unfavorable prognosis.

All models underwent 5000 permutation tests to assess their reliability. The DeLong test was used to compare the statistical performance of models. Decision curve analysis (DCA) was conducted on the best predictive model to evaluate its clinical value. Additionally, Shapley Additive exPlanations (SHAP) were employed to determine the contribution of features to the predictive performance of the model.

Statistical analysis

SPSS software (Version 26.0, IBM, Armonk, NY, USA) was used for statistical analysis. Normality and homogeneity of variance tests were performed on the continuous clinical data. Normally distributed data are presented as mean ± standard deviation, while non-normally distributed data are presented as median (first quartile, third quartile). Continuous variables were analyzed using either independent samples t-test or Mann–Whitney U test. Categorical variables were expressed as n (%), and inter-group comparisons were performed using the chi-square test. Image processing and model construction were performed using MATLAB R2022b (http://www.mathworks.com) and Python (http://www.python.org, version 3.11). Statistical significance was set at $P<0.05$.

Results

Clinic characteristics

The demographic and clinical data for the AIS patients included are specified in Table 1. A total of 84 patients had a favorable prognosis, with 26 having an unfavorable prognosis. Patients with an unfavorable prognosis were older ($P=0.044$) and had a higher baseline NIHSS score ($P<0.001$) compared to those with a favorable prognosis. No significant statistical differences were observed between the two groups regarding gender, onset to therapy time, treatment method, history of atrial fibrillation, hypertension, diabetes, hyperlipidemia, coronary artery disease, smoking, alcohol abuse, and history of stroke.

Model performance and validation

The AUC of the infarct radiomics models established using five ML algorithms ranged from 0.569 to 0.707 (Fig. 2A), the whole-brain radiomics models had an AUC ranging from 0.598 to 0.738 (Fig. 2B), and the combined radiomics models showed an AUC ranging from 0.707 to 0.772 (Fig. 2C). The DeLong test showed that the combined radiomics models significantly enhanced predictive performance with RF ($P=0.010$), XGBoost ($P=0.002$), and LightGBM ($P=0.004$) algorithms compared to the infarct radiomics models, while LR ($P=0.585$) and SVM ($P=0.635$) algorithms did not show significant improvement. Furthermore, no significant differences in performance were found between the infarct and the whole-brain radiomics models across all algorithms ($P=0.108$–0.992). Moreover, there was no significant difference in performance between the combined radiomics models and the whole-brain radiomics models ($P=0.156$–0.781) or the clinical models ($P=0.120$–0.671) across all algorithms.

The AUC of the clinical models ranged from 0.645 to 0.792 (Fig. 2D), while the AUC of the comprehensive models ranged from 0.832 to 0.904 (Fig. 2E). The DeLong test results indicated that the comprehensive models outperformed both the clinical models and the combined radiomics models significantly in terms of predictive performance across all algorithms (LR: $P=0.039$, 0.001; SVM: $P=0.014$, 0.001; RF: $P=0.010$, 0.001; XGBoost: $P=0.001$, 0.003; LightGBM: $P=0.001$, 0.005).

Table 2 contains the detailed metrics for all models, showing high reliability in the 5000 permutation tests for each model. The best performance was achieved by the comprehensive model utilizing the SVM algorithm, with an AUC of 0.904. Moreover, it achieved a precision-recall AUC of 0.766 in predicting unfavorable prognosis. Grid search determined that the optimal hyperparameter settings for the SVM algorithm included a linear kernel function and a penalty parameter of 1.0.

Table 1 Demographic and clinical characteristics of the AIS patients

Characteristics	Favorable ($n=84$)	Unfavorable ($n=26$)	t/Z/χ^2	*P* value
Age, year[a]	66.37 ± 12.02	71.92 ± 12.56	-2.038	0.044*
Female, n (%)	33(39.29)	14(53.85)	1.720	0.190
Onset to therapy time, hour[a]	12.91 ± 14.55	12.92 ± 13.50	-0.004	0.997
Baseline NIHSS score[b]	3(1, 5.75)	8(4, 11.25)	4.763	< 0.001*
Reperfusion therapy, n (%)	21(25.00)	7(26.92)	0.039	0.844
Atrial fibrillation, n (%)	13(15.48)	5(19.23)	0.205	0.651
Hypertension, n (%)	62(73.81)	20(76.92)	0.101	0.750
Diabetes, n (%)	23(27.38)	8(30.77)	0.113	0.737
Hyperlipidemia, n (%)	18(21.43)	6(23.08)	0.032	0.859
Coronary heart disease, n (%)	3(3.57)	2(7.69)	0.777	0.378
Smoking, n (%)	20(23.81)	4(15.38)	0.826	0.363
Alcohol abuse, n (%)	13(15.48)	3(11.54)	0.248	0.619
History of stroke, n (%)	10(11.90)	7(26.92)	3.427	0.064

n, number of cases

[a] ($\bar{x} \pm s$)

[b] [M (Q1, Q3)]

NIHSS: National Institutes of Health Stroke Scale

The marked with "*" is statistically significant ($P<0.05$)

Decision curve analysis

We generated DCA curves for the comprehensive model, clinical model, and combined radiomics model using the SVM algorithm. The comprehensive model showed a higher net clinical benefit across the entire threshold range when compared to the other two models, as depicted in Fig. 3.

Model interpretability

Figure 4 illustrates the SHAP analysis of the SVM comprehensive model. The baseline NIHSS score was identified as the most crucial factor in predicting AIS prognosis, with age also playing a role in the prediction. Furthermore, infarct shape features, high-order texture and statistical features of both the infarct and whole-brain significantly enhanced the model's predictive performance.

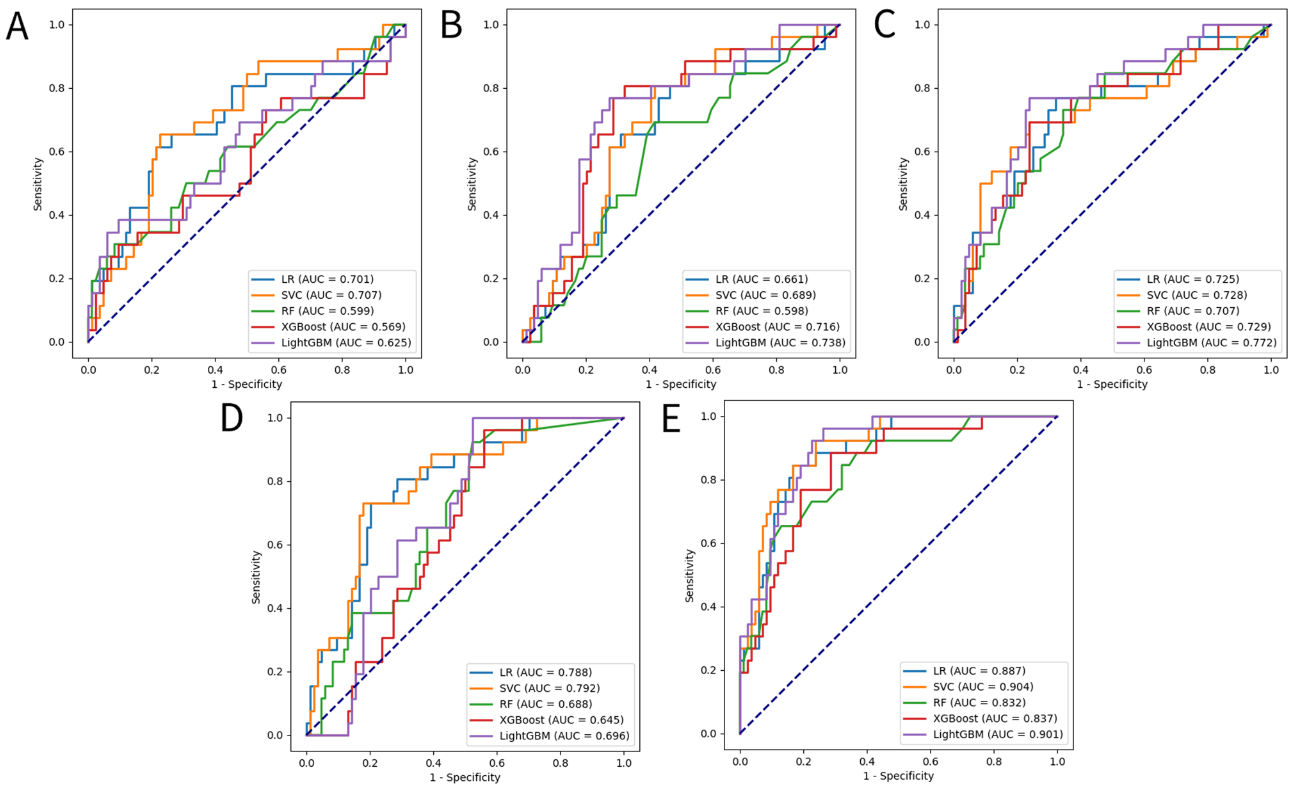

Fig. 2 Receiver operating characteristic curve of models in predicting prognosis of AIS. (**A**) infarct radiomics models. (**B**) whole-brain radiomics models. (**C**) combined radiomics models. (**D**) clinical models. (**E**) comprehensive models. LR: logistic regression; SVM: support vector machine; RF: random forest; XGBoost: extreme gradient boosting; LightGBM: light gradient boosting machine

Discussion

This study extracts radiomics features from infarct and whole-brain multi-PLD ASL CBF images in AIS patients. The prediction models for AIS prognosis were developed by combining clinical data and utilizing five ML algorithms. The findings suggest that utilizing CBF imaging radiomics features from multi-PLD ASL can effectively improve the predictive accuracy of clinical risk factors for the prognosis of stroke patients.

In our study, the AUC of the whole-brain radiomics models ranged from 0.598 to 0.738, while that of the infarct radiomics models ranged from 0.569 to 0.707. The combined radiomics model, incorporating both, was constructed with an AUC ranging from 0.707 to 0.772. When using the RF, XGBoost, and LightGBM algorithms, the predictive performance of the combined radiomics models weas significantly higher than that of the infarct radiomics models. Our results further support Guo et al.'s findings, suggesting that whole-brain perfusion radiomics features can partially predict AIS prognosis and that combining them with infarct radiomics features improves prediction accuracy [24]. Moreover, we observed that the combined radiomics models tended to outperform the whole-brain radiomics models. Although the DeLong test did not show a significant difference between the two, this could be attributed to the small sample size, leading to inadequate statistical power [27]. Consequently, there is still justification to suggest that the combined radiomics models are better than the whole-brain radiomics models. Notably, the performance of the whole-brain radiomics models was comparable to that of the infarct radiomics models. Nevertheless, the whole-brain radiomics model does not rely on infarct segmentation, which simplifies the prognosis assessment process, particularly in cases with unclear or complex infarct boundaries.

When analyzing the clinical data of AIS patients, it was observed that individuals with unfavorable prognoses tended to have higher ages and baseline NIHSS scores, which are commonly recognized as independent risk factors for prognosis [28, 29]. The AUC of the clinical models developed from clinical data ranged from 0.645 to 0.792, showing similarity to the combined radiomics models. However, depending only on clinical data for prognosis prediction has some constraints, especially when clinical data cannot be promptly, precisely, or thoroughly obtained. The combined radiomics models, based on early-stage images, can quickly provide clinicians with accurate prognosis assessments, offering a potential

Table 2 The performance of models in predicting prognosis of AIS patients

Models	AUC (95% CI)	Sensitivity	Specificity	Accuracy	F1 score	Permutation test
Infarct radiomics						
LR	0.701 (0.683–0.742)	0.615	0.786	0.745	0.533	< 0.001*
SVM	0.707 (0.674–0.772)	0.654	0.774	0.745	0.548	< 0.001*
RF	0.599 (0.556–0.623)	0.308	0.917	0.773	0.390	0.002*
XGBoost	0.569 (0.542–0.580)	0.308	0.905	0.764	0.381	0.002*
LightGBM	0.625 (0.615–0.629)	0.385	0.905	0.782	0.455	0.002*
Whole-brain radiomics						
LR	0.661 (0.641–0.682)	0.654	0.69	0.682	0.493	0.002*
SVM	0.689 (0.675–0.704)	0.808	0.583	0.636	0.512	0.002*
RF	0.598 (0.567–0.620)	0.692	0.583	0.609	0.456	0.003*
XGBoost	0.716 (0.700–0.736)	0.808	0.679	0.709	0.568	< 0.001*
LightGBM	0.738 (0.719–0.758)	0.769	0.726	0.736	0.580	< 0.001*
Combined radiomics						
LR	0.725 (0.682–0.749)	0.769	0.679	0.700	0.548	< 0.001*
SVM	0.728 (0.691–0.736)	0.692	0.762	0.745	0.562	< 0.001*
RF	0.707 (0.689–0.767)	0.731	0.655	0.673	0.514	< 0.001*
XGBoost	0.729 (0.685–0.749)	0.692	0.762	0.745	0.562	< 0.001*
LightGBM	0.772 (0.745–0.804)	0.769	0.762	0.764	0.606	< 0.001*
Clinical						
LR	0.788 (0.693–0.871)	0.731	0.798	0.782	0.613	< 0.001*
SVM	0.792 (0.694–0.878)	0.731	0.821	0.800	0.633	< 0.001*
RF	0.688 (0.576–0.785)	0.923	0.476	0.582	0.511	0.001*
XGBoost	0.645 (0.539–0.742)	0.962	0.440	0.564	0.510	0.001*
LightGBM	0.696 (0.593–0.787)	1.000	0.476	0.600	0.542	< 0.001*
Comprehensive						
LR	0.887 (0.815–0.948)	0.846	0.833	0.836	0.710	< 0.001*
SVM	0.904 (0.840–0.959)	0.923	0.762	0.800	0.686	< 0.001*
RF	0.832 (0.734–0.919)	0.846	0.679	0.718	0.587	< 0.001*
XGBoost	0.837 (0.745–0.917)	0.885	0.714	0.755	0.630	< 0.001*
LightGBM	0.901 (0.837–0.953)	0.962	0.738	0.791	0.685	< 0.001*

AUC: area under the curve; CI: confidence interval; LR: logistic regression; SVM: support vector machine; RF: random forest; XGBoost: extreme gradient boosting; LightGBM: light gradient boosting machine. The marked with "*" is statistically significant ($P < 0.05$)

alternative to clinical models and proving valuable for patients lacking clinical data.

This study further evaluated the efficacy of integrating CBF radiomics features with clinical data. The comprehensive models significantly improved the accuracy of AIS prognosis prediction, with an AUC range of 0.832 to 0.904. The DeLong test indicated that they surpassed both the clinical models and the combined radiomics models across all ML algorithms. It appears that both clinical data and CBF radiomics features have limited prognostic value when employed individually. In contrast, the comprehensive models combine the value of CBF radiomics features, reflecting changes in brain perfusion, with clinical data, which reflects the patient's overall health, providing a more accurate prognosis for AIS patients. This discovery aligns with multiple recent radiomics studies, emphasizing the significance of both radiomics features and clinical data in predicting AIS prognosis [30].

The comprehensive model created using the SVM algorithm demonstrated the highest performance, with an AUC of 0.904, surpassing the XGBoost algorithm model from our previous study with an AUC of 0.876 [16]. Additionally, the model demonstrates outstanding performance for predicting unfavorable prognosis. The SVM algorithm has the potential to improve the accuracy of prediction by adjusting the classification threshold and optimizing the decision boundary, especially in situations with imbalanced classification categories [31]. Therefore, the SVM algorithm demonstrates superior predictive performance compared to several other algorithms. Although this model is still slightly inferior to the model developed by Guo et al. using DSC-PWI technology (AUC = 0.971), the multi-PLD ASL technology used in this study overcomes the limitation of DSC-PWI, which relies on contrast agents [24]. This is highly important for patients who have contraindications to contrast agents.

The DCA confirmed the clinical significance of the SVM-based models, with the comprehensive model

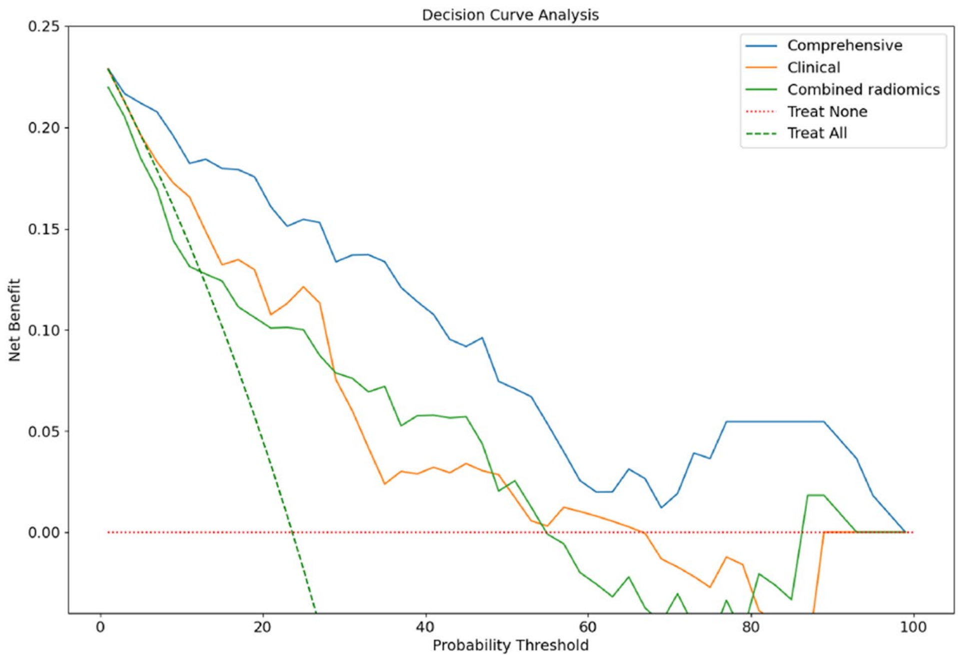

Fig. 3 Decision curve analysis of the SVM models

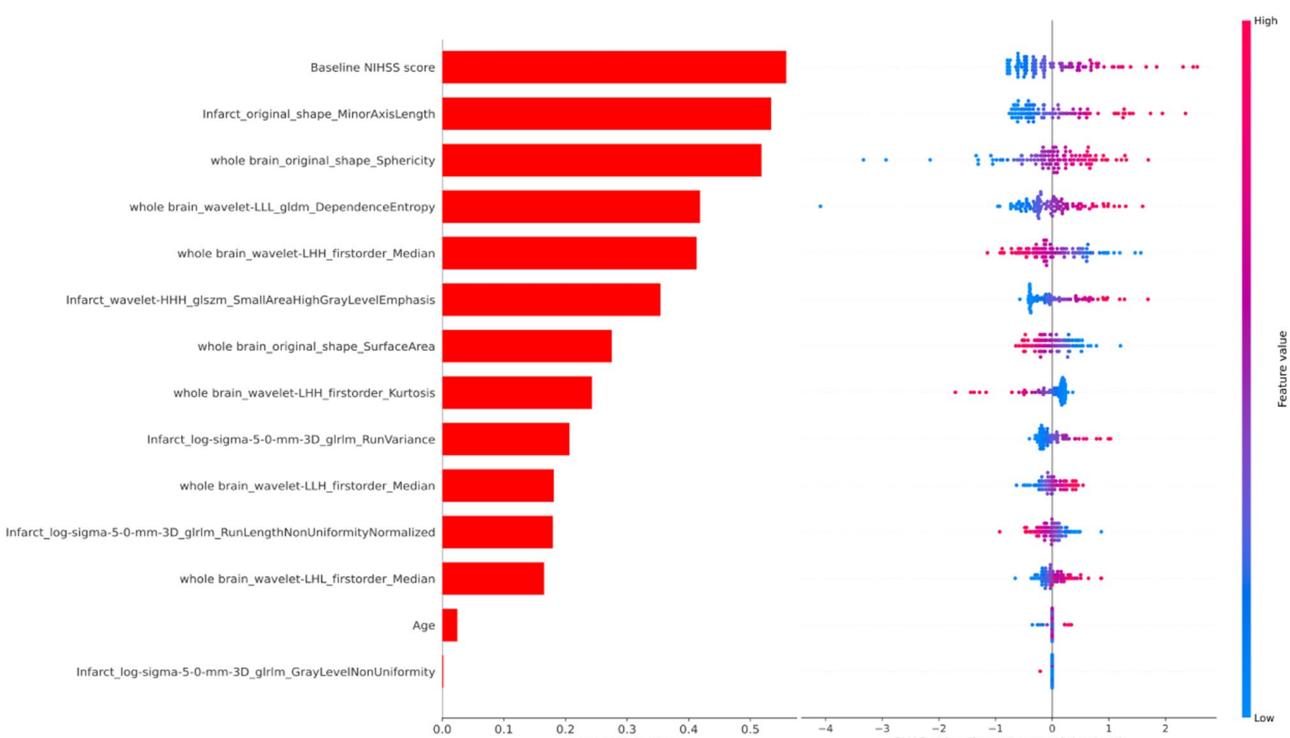

Fig. 4 The SHAP values of the SVM comprehensive model. NIHSS: national institutes of health stroke scale

displaying a higher net benefit than the clinical or combined radiomics model across all thresholds. This indicates that the SVM comprehensive model might offer clinicians more accurate prognostic assessments at different risk thresholds. In future clinical applications, clinicians will be able to quickly assess AIS patients' prognosis through the combination of the patient's imaging data and basic clinical information.

We performed SHAP analysis on the SVM comprehensive model to assess the contribution of each feature to predictive accuracy and to identify key factors influencing AIS prognosis. The baseline NIHSS score emerged as the most significant predictor, in accordance with several prior studies [32]. The severity of neurological dysfunction in AIS patients is often reflected by a higher baseline NIHSS score, which is closely tied to clinical prognosis [33]. Furthermore, age was identified as a critical factor in AIS prognosis. As age increases, neural plasticity gradually declines, reducing the brain's ability to recover from ischemic injury [34]. Ahmed et al. confirmed that older AIS patients have a significantly higher prevalence of complications, further increasing the risk of unfavorable prognosis [35]. Of the CBF radiomics features, the infarct shape features were the most significant. They often reflect the extent of ischemic damage, with larger or irregular infarcts indicating more severe vascular occlusion or collateral failure, both of which affect the prognosis of AIS [36]. Furthermore, higher-order statistical and texture features from both whole-brain and infarct also played crucial roles in the model. The higher-order statistical features of CBF images provide multidimensional quantification of perfusion states, aiding in differentiating the ischemic penumbra from the infarct core and identifying AIS patients with potentially reversible ischemic damage [37]. High-order texture features evaluate complex patterns of pixel intensity and spatial arrangement, enabling the detection of subtle brain perfusion abnormalities that may be imperceptible through visual inspection alone [38]. Additionally, capillary rupture caused by AIS can result in heightened irregularity in imaging texture patterns [39]. In summary, the baseline NIHSS score, age, infarct shape features, and higher-order statistical and texture features from both whole-brain and infarct in CBF imaging are key factors influencing AIS prognosis. These factors collectively improve the predictive accuracy of the SVM comprehensive model.

This study has several limitations. First, the sample size is relatively small, which poses a limitation on the robustness of the findings. Although leave-one-out cross-validation and permutation tests were employed to mitigate this issue, future studies should validate the model's reliability using larger datasets. Second, the exclusion of posterior circulation AIS cases limits generalizability; future studies should include these cases to improve model applicability. Third, the variable MRI timing (24–72 h) may introduce temporal heterogeneity in radiomics features, and future studies should stratify or adjust for scan time to enhance model robustness. Fourth, although ICC analysis was performed to ensure consistency in manual infarct segmentation, inter-observer variability may still affect result accuracy. The absence of standardized DWI lesion segmentation also prevented the inclusion of DWI–CBF mismatch analysis, a meaningful marker of the ischemic penumbra. Future studies should adopt automated or semi-automated segmentation methods to reduce bias and enable reliable mismatch assessment. Finally, this study primarily focused on the multi-PLD ASL sequence. Although this method offers significant advantages, it is recommended that future research delve deeper into multimodal imaging, including diffusion tensor imaging and susceptibility-weighted imaging. These modalities may provide additional information on brain tissue damage and ultimately improve the accuracy of prognosis prediction for AIS.

Conclusions

The whole-brain CBF radiomics features of multi-PLD ASL can replace the infarct radiomics features that need to be delineated. The performance of the combined radiomics model has been improved to a certain extent compared to the infarct radiomics model, equivalent to the clinical model, which is applicable to stroke patients with incomplete clinical data. The CBF radiomics features from multi-PLD ASL can effectively improve the predictive accuracy of clinical risk factors for the prognosis of stroke patients. The comprehensive model integrates multi-PLD ASL CBF radiomics and clinical data, providing a safe and accurate tool for early prognosis prediction in AIS patients.

Abbreviations

AIS	Acute ischemic stroke
CBF	Cerebral blood flow
DSC-PWI	Dynamic sensitivity contrast perfusion weighted imaging
ASL	Arterial spin labeling
MRI	Magnetic resonance imaging
PLD	Post-labeling delay
ATT	Arterial transit time
mRS	Modified Rankin Scale
NIHSS	National Institutes of Health Stroke Scale
3D-T1WI	Three-dimensional T1-weighted imaging
DWI	Diffusion weighted imaging
ROI	Region of interest
ICC	Interclass correlation coefficient
ML	Machine learning
LR	Logistic regression
SVM	Support vector machine
RF	Random forest
XGBoost	Extreme gradient boosting
LightGBM	Light gradient boosting machine
AUC	Area under the curve
DCA	Decision curve analysis
SHAP	Shapley Additive exPlanations

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None.

Author contributions

ZYW: data analysis and manuscript writing. CJJ: project development. XXZ: data collection and data analysis. TCM: methodology and software. QQL and SW: data collection. CSD and YS: investigation and manuscript editing. ZYD and FC: project development, funding acquisition and manuscript editing. All authors read and approved the final manuscript.

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Data availability
The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
This study was conducted according to the tenets of the Declaration of Helsinki. Informed consent was obtained from each patient. The study was approved by the Ethics Committee of the Yancheng Third People's Hospital (Ethics Approval No. 2020-77).

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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